1. Targets and general care

BTF 4th ed / SIBICC

- Treat ICP > 22 mmHg; CPP 60–70 mmHg
- SBP ≥ 100 (50–69 yr) / ≥ 110 (15–49 or > 70)
- Avoid hypoxia (SpO2 > 94%, PaO2 10–13 kPa), PaCO2 4.5–5 kPa
- Na 140–145, glucose 6–10, normothermia, Hb > 70 (TRAIN: > 90 better in brain injury)

Who needs ICP monitoring

- GCS 3–8 with abnormal CT, or normal CT + 2 of age > 40, posturing, SBP < 90

Key point: Prevent secondary injury: hypoxia, hypotension, hyper/hypocapnia, fever, hyponatraemia.

2. Tier 0 and tier 1

Tier 0 (all)

- Head-up 30°, neck neutral, loose ties, sedation/analgesia, normothermia, seizures prophylaxis 7 days (levetiracetam/phenytoin)

Tier 1

- CPP 60–70, ↑ sedation/analgesia, PaCO2 4.7–5.3 kPa
- Osmotherapy: mannitol 0.25–1 g/kg or hypertonic saline (Na up to 155, osm < 320)
- CSF drainage if EVD, EEG

Key point: Tiers: escalate only when the lower tier fails; go back down when ICP settles.

3. Tier 2 and tier 3 (SIBICC)

Tier 2

- Mild hypocapnia 4.0–4.5 kPa (with PbtO2/SjvO2 monitoring)
- Neuromuscular blockade trial
- MAP challenge to assess autoregulation
- Repeat CT, consult neurosurgery

Tier 3

- Barbiturate coma (titrate to burst suppression)
- Secondary decompressive craniectomy
- Mild hypothermia 35–36 °C

Key point: Know what belongs to each tier: examiners ask exactly that.

4. Evidence for rescue therapies

Hyperventilation

- ↓ ICP by vasoconstriction but ↓ CBF → ischaemia; avoid $PaCO_2 < 4$ and in first 24 h

Barbiturates

- ↓ ICP, no outcome benefit; hypotension, infection, ileus

Decompressive craniectomy

- **DECRA 2011**: early bifrontal → worse outcome
- **RESCUEicp 2016**: last tier ICP > 25 → ↓ mortality (27 vs 49%) but ↑ vegetative/severe disability

Hypothermia

- **Eurotherm3235**: ICP-driven hypothermia ↑ unfavourable outcome
- **POLAR**: prophylactic hypothermia no benefit

Key point: Lower ICP ≠ better outcome: the classic exam point.

5. Advanced neuromonitoring

PbtO2

- Target > 20 mmHg; treat hypoxia via FiO2, CPP, PaCO2, Hb, sedation
- BOOST-II: ↓ time with hypoxia, trend to better outcome; BOOST-3 ongoing/results

Microdialysis

- Lactate/pyruvate ratio > 25–40 = metabolic crisis; glucose, glutamate

Others

- PRx (autoregulation) → CPPopt; SjvO2 55–75%; TCD; continuous EEG; pupillometry

Key point: Multimodal monitoring personalises CPP and PaCO2 targets.

6. Other TBI essentials

Points

- TXA within 3 h in mild-moderate TBI (CRASH-3)
- No steroids (CRASH: harm)
- Albumin harmful (SAFE-TBI)
- VTE prophylaxis after stable CT (24–72 h)
- Prognosis: IMPACT/CRASH calculators; avoid early withdrawal

Key point: CRASH-3 for TXA, CRASH for steroids, SAFE-TBI for albumin.